

## Week 4 problems (NPTEL)

Q.384) The quality control manager at Li-Battery factory needs to determine whether the mean life of a large shipment of Li-Battery is equal to the specified value of 375 hours. The process std. deviation is known to be 100 hours. A random sample of 64 batteries indicates a sample mean life of 350 hrs. State the null hypothesis, at the  $\alpha = 0.05$  level of significance, is there any evidence that mean life differs from 375 hours.

sol<sup>n</sup>

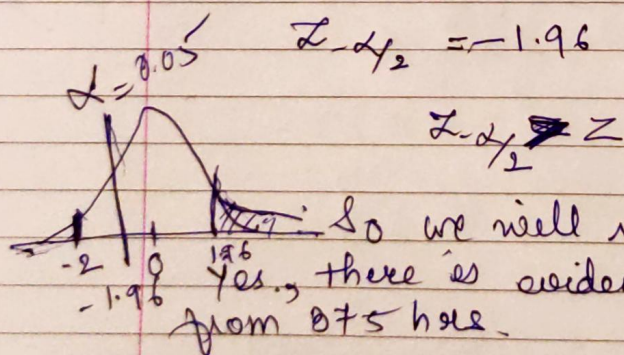
$$\mu = 375, \sigma = 100$$

$$\bar{x} = 350, n = 64$$

null hypothesis:  $\mu = 375$

Alternate hypothesis =  $\mu \neq 375$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{350 - 375}{100/\sqrt{64}} = \frac{-25}{100/8} = -2$$



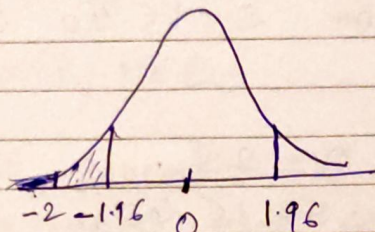
5) In above question p-value is

for  $Z = -2$ , p-value = 0.01831

$$P(Z < -2) = 1 - 0.98169$$



p-value - It is the probability for the null hypothesis to be true



Area at  $z = 2 \rightarrow 0.0228$

$$p\text{-value} = 2 \times 0.0228 = 0.0456$$

$$p\text{-value} < \alpha$$

6) In question 3, at a 95% confidence interval, the estimate of population mean life of the battery is —

95% confidence interval,

$$\text{Lower limit} = \bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$= 350 - 1.96 \cdot \frac{100}{\sqrt{64}}$$

$$= 350 - 24.5 = 325.5$$

Upper limit is

$$\bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$350 + 24.5 = 374.5$$

$\mu = 375$  doesn't lie in the interval 325.5 to 374.5 so we will reject null hypothesis



4) The mean cost of a hotel room in a city is \$168 per night. A random sample of 25 hotels resulted in  $\bar{x} = \$172.50$ . Sample standard deviation  $S = 15.40$ . Calculate the t-statistic.

soln

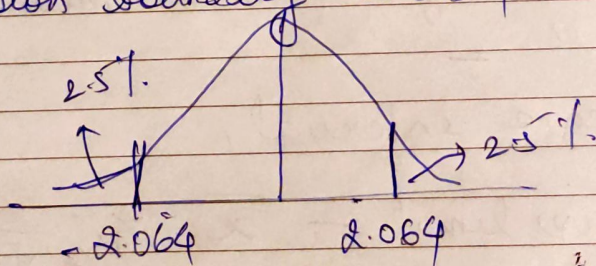
$$\mu = 168, n = 25$$

$$\bar{x} = 172.50, S = 15.40$$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{172.50 - 168}{15.40/\sqrt{25}} = \underline{\underline{1.461}}$$

$$t_{\alpha/2} \text{ degree of freedom} = n - 1 = 24$$

Decision Boundary 2.064



$$t < t_{\alpha/2}$$

$\therefore$  Do not reject the null hypothesis

$\therefore$  Mean of the hotel room price is 168%



Week 5 - solution

- 3) The 'F' ratio of in a completely randomized ANOVA is the ratio of

sol<sup>n</sup>  $F = \text{MSTR} / \text{MSE}$

- 4) An ANOVA procedure is applied to data obtained from 6 samples where each sample contains 10 observations. The degrees of freedom for critical value of F are

sol<sup>n</sup> Degrees of freedom for numerator (SSB) is  $K - 1 = 5$  where K is no. of groups.

Degrees of freedom for denominator is  $N - K$  where N is total no. of observations

Here  $\text{dof} = 60 - 6 = 54$

So answer is

5 numerator & 54 denominator degrees of freedom.

- 5) In an ANOVA problem if  $\text{SST} = 120$ ,  $\text{SSTR} = 80$  then SSE is

$$\text{SST} = \text{SSTR} + \text{SSE}$$

$$\text{SSE} = \text{SST} - \text{SSTR} = 120 - 80 = 40$$

- 6) The critical F value with 8 numerator & 29 denominator degrees of freedom at  $\alpha = 0.01$  is

sol<sup>n</sup>

from F-table

critical F-value is 3.20



- 7) Two independent simple random samples are taken to test the difference beto the means of 2 populations. The std deviations are not known but are assumed to be equal. The sample sizes are  $n_1 = 15$ , &  $n_2 = 35$ . The correct distribution to use is the  $t$  distribution with 48 degrees of freedom

Soln

$$dof = n_1 + n_2 - 2 = 15 + 35 - 2 = 48$$

- 8) Mean marks obtained by male & female students of schools ABCD in the first unit test are shown below

|                                    | Male | Female |
|------------------------------------|------|--------|
| sample size                        | 64   | 36     |
| sample mean marks                  | 44   | 41     |
| Population variance ( $\sigma^2$ ) | 128  | 72     |

The std. error of diff. betn 2 means is

Soln

$$\begin{aligned}
 SE &= \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2} \\
 &= \sqrt{128/64 + 72/36} \\
 &= \sqrt{2 + 2} = \sqrt{\frac{128}{64} + \frac{72}{36}} \\
 &= \sqrt{2+2} \\
 &= \sqrt{4} = 2
 \end{aligned}$$



- 10) Refer to Q. 9. If you are interested in testing whether or not the average marks of males are significantly greater than that of females, the test statistic is

Soln

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{44 - 41}{2} = \frac{3}{2} = 1.5$$



## Week-6 solution

Q.3) A regression analysis between sales ( $Y$  in \$1000) & advertising ( $X$  in dollars) resulted in the following eq<sup>n</sup>.

$$Y = 30,000 + 5X$$

The above eq<sup>n</sup> implies that an

sol<sup>n</sup> Increase in \$1 in advertising is associated with an increase of \$5000 in sales.

Q.4) In a regression & correlation analysis if  $r^2$  square = 1 then

$$r^2 = \frac{SSR}{SST}$$

$$SSR = SST$$

$$\therefore SST = SSR + SSE$$

Q.5  $SSE$  will be zero.

Q.5  $SSE$  can never be larger than SST

Q.6. For the given data determine the R-squared value.

Data

Miles travel

Petrol consumption in litre

20

1

45

3

56

5

34

2

28

1.6

49

3.7



Week 7

Q For a multiple regression model with 2 independent variable  $R^2 = 0.904$ , & adjusted  $R^2 = 0.88$ , determine the number of observations

Soln

$$\text{Adj } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where  $p$  is no. of independent variables = 2

$$0.88 = 1 - \frac{(1 - 0.904)(n - 1)}{n - 3}$$

$$0.88(n - 3) = n - 3 - [n + 0.904 - 0.904n - 1]$$

$$0.88n = 2.64 - \cancel{n} - 3 - \cancel{n} - 0.904 + 0.904n + 1$$

$$0.88n - 0.904n = 2.64 - 3 - 0.904 + 1$$

$$(-0.024)n = -0.264$$

$$n = \underline{\underline{11}}$$